

Structured Output and Schemas

Project: Verified Study Buddy

1. Your mission

Your learning target

I can separate JSON parsing, schema validation and source support checks.

I can reject invalid types and unexpected fields under a closed schema.

I can trace a validly formatted but unsupported claim.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

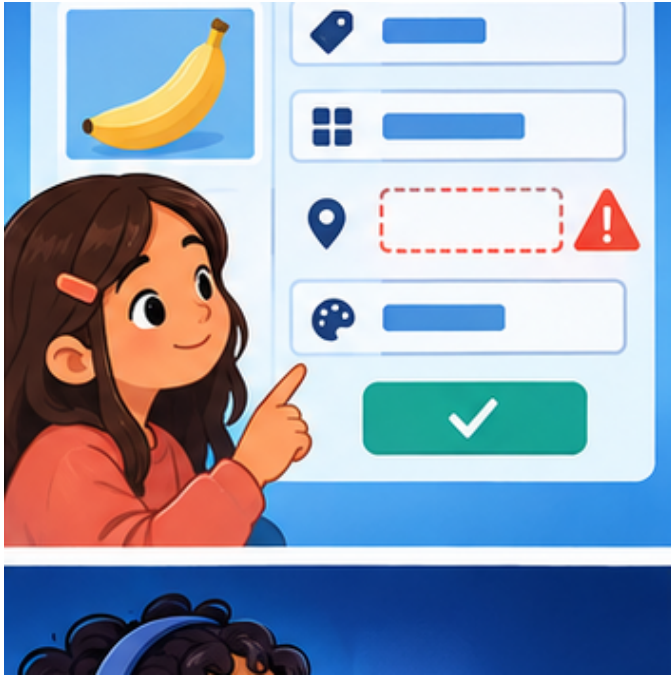
Without opening the old lesson, explain one key idea from Prompt as a Contract. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from A Distribution of Possibilities. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 8: Structured Output and Schemas

Schemas make output easier to validate, reuse, and test.



CREATOR MISSION

Request fields, then check missing and invalid values.

Creator name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Does a requested output format prove a response followed the specification?

This week's end goal

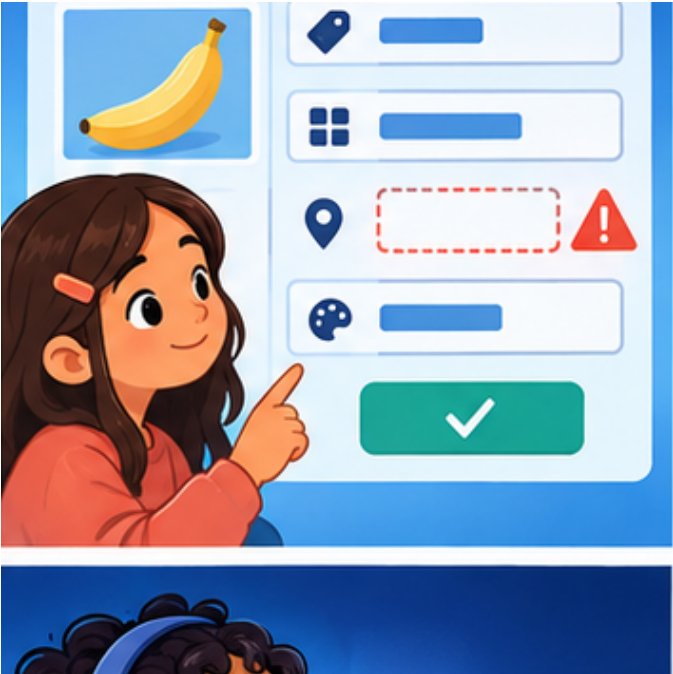
By the end, I can validate structured output in layers: syntax, schema and source meaning.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use structured output and schemas to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

A response may be valid JSON and still violate the schema or sources. First parse the text. Then check the exact object fields and types. Finally check source support and domain meaning. These checks answer different questions. Do not call a successful parser a complete validator.

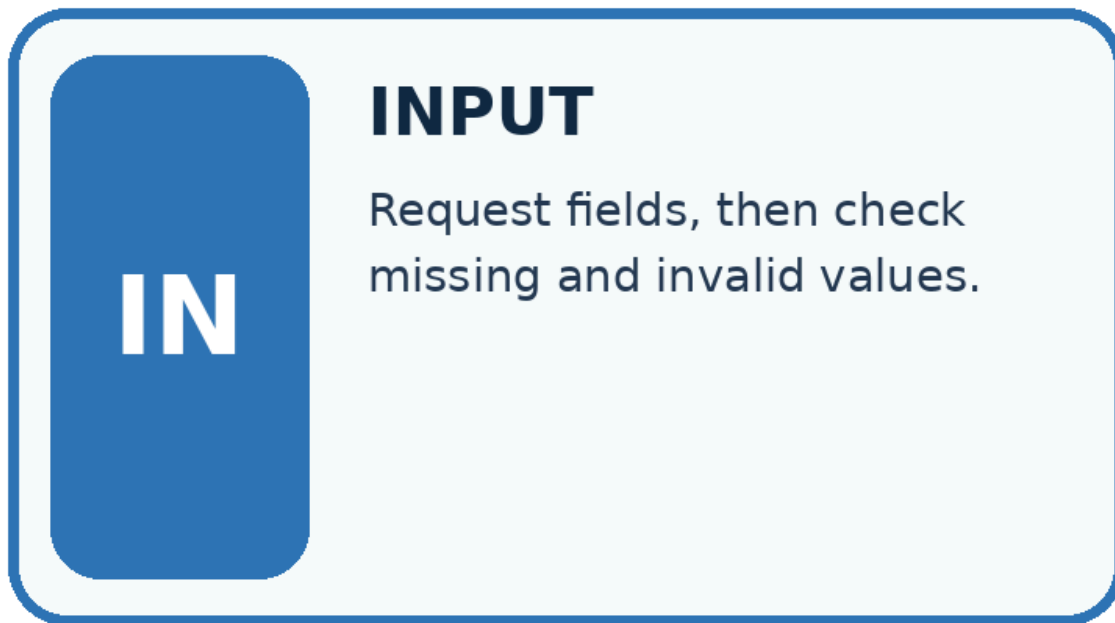
Use the Week 11 closed object: exactly status,room,capacity,source_ids. status is supported or needs_review. source_ids is a list of distinct known source-ID strings. For supported, room is a nonempty string and capacity a nonnegative integer. For needs_review, room and capacity are null. Boolean values are not capacities under this contract.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



SYSTEM

Schemas make output easier to validate, reuse, and test.





EVIDENCE

Request fields, then check missing and invalid values.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

Response fragment · Layer that rejects it · Reason

Response fragment: {status: supported}

Layer that rejects it: Syntax

Reason: JSON keys/strings need quotes

Response fragment: {"capacity": "12"}

Layer that rejects it: Schema in a complete object

Reason: String does not satisfy integer capacity

Response fragment: {"capacity": true}

Layer that rejects it: Schema in a complete object

Reason: Boolean is not accepted as an integer

Response fragment: An extra booking field

Layer that rejects it: Closed schema

Reason: Unexpected field

These fragments illustrate failures, not complete candidate objects. A real check receives the whole object and verifies every required field. Do not repair a string capacity automatically unless a separate documented conversion policy permits it; otherwise report validation failure. A null value differs from missing a field.

Keep the evidence roles separate

Plan or input · Procedure · Permitted conclusion

Plan or input: Generated text

Procedure: JSON parse

Permitted conclusion: A value or syntax error

Plan or input: Parsed value

Procedure: Closed-schema check

Permitted conclusion: Contract match or schema error

Plan or input: Contract-valid object

Procedure: Match claims to current sources

Permitted conclusion: Supported or semantic error

Each passed layer still leaves later layers to check.

Connect the picture and the explanation

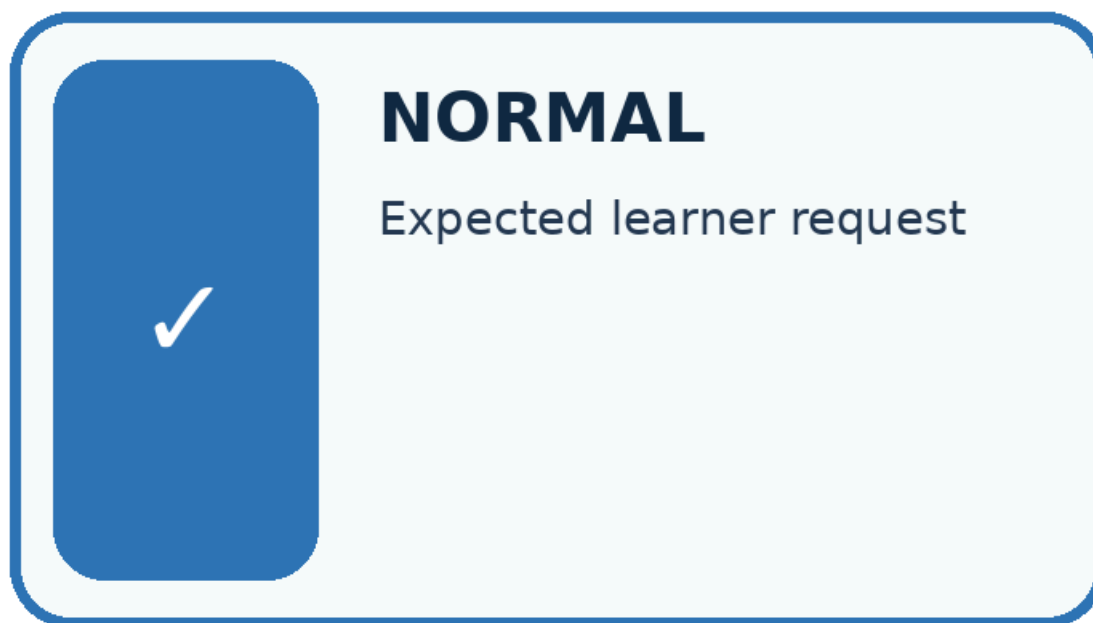
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.





AMBIGUOUS

Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING SOURCE

Evidence unavailable



MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: Add an extra field "rating":5 to X under the exactly-fields schema.

Prediction: The changed object fails schema despite keeping the correct page count.

Reason: Exactly means additional fields are not allowed in this contract.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

SCHEMA

required fields and value rules

2

FIELD

named piece of data

3

TYPE

kind of value allowed

4

VALIDATION

check output against rules

Word · My own example · Not this / boundary

Word: schema

My own example: A specification of required fields, types and structural rules for data.

Not this / boundary: _____

Word: field

My own example: A named item of information in a record or structured message.

Not this / boundary: _____

Word: type

My own example: The category of a value, such as text, number or Boolean.

Not this / boundary: _____

Word: validation

My own example: Checking data or behaviour against rules; in ML, validation data is used to choose settings or models, separately from final testing.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Syntax validation

Meaning: Checking whether text can be parsed under the format grammar

Word: Schema validation

Meaning: Checking fields, types and allowed combinations in a parsed value

Word: Semantic validation

Meaning: Checking meaning and consistency with sources or domain rules

Word: Closed schema

Meaning: A contract rejecting fields beyond the stated set

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 8 • WORKED EVIDENCE

- 1 Schema fields: answer=text; evidence=list; confidence=low/medium/high.
- 2 Output omits evidence: validation fails.
- 3 Confidence=97 is the wrong type and allowed-value set.
- 4 Reject, repair, or request again before downstream use.

1. Schema fields: answer=text; evidence=list; confidence=low/medium/high.
2. Output omits evidence: validation fails.
3. Confidence=97 is the wrong type and allowed-value set.
4. Reject, repair, or request again before downstream use.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Suppose sources P and Q approve room B and give capacity 12. The complete object `{"status":"supported","room":"B","capacity":99,"source_ids":["P","Q"]}` parses and satisfies the stated field types, but its capacity is unsupported by Q. Reject it at semantic validation. Listing a real source ID does not prove that source supports the associated claim.

An object with `needs_review` and room B violates the conditional schema rule even if its source IDs are real. An object with `supported`, room B and capacity 12 but source ID Z fails the known-source rule. Require the linked current approval and capacity records to support the same room.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Structured Output and Schemas

A fictional Study Buddy returns a machine-readable book fact.

The facts we will use

Schema: exactly title text, pages integer, source_id text. Approved record R: River Notes, 64 pages.

$X = \{\text{"title": "River Notes", "pages": 64, "source_id": "R"}\}$; Y uses pages: "64"; Z uses pages: 46 with otherwise identical fields.

Structured Output and Schemas

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Parse

Each shown object is valid JSON syntax.



2. Schema check

Y fails because its pages value is text, not an integer.



3. Meaning check

Z fits field types but contradicts R's page count.



4. Accept X

X passes syntax, declared schema and source comparison.

CHANGE ONE THING

Add an extra field "rating":5 to X under the exactly-fields schema.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Parse

What happens and why: Each shown object is valid JSON syntax.

Step: 2. Schema check

What happens and why: Y fails because its pages value is text, not an integer.

Step: 3. Meaning check

What happens and why: Z fits field types but contradicts R's page count.

Step: 4. Accept X

What happens and why: X passes syntax, declared schema and source comparison.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Parsing establishes that text forms a structured object. Schema checks establish required fields and types. Semantic validation compares their meaning with the approved source. These are different layers: a parseable, well-shaped object can still contain a false fact. Report the first failing layer without treating format success as factual success.

What this example does not establish

Source agreement does not establish that the approved catalogue itself is correct or complete.

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: Valid-looking braces prove the content is correct. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Keep validation outcomes and the original generated text for fictional debugging data. Report the first failing layer and enough detail to locate the issue; later validation should not be reported as passed when it was never reached. Even a fully valid reporting object does not authorise a booking. The application retains the read-only permission boundary.

Exit: a parser accepts JSON, but capacity is true and an extra book_now field is present. State which checks remain and why no booking should occur.

8. Practise together

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Supported object {"status":"supported","room":"D","capacity":8,"source_ids":["P","Q"]} matches approval P for D and register Q for D capacity 8. It passes all three layers. Replacing capacity with 0 still passes schema because zero is nonnegative, but fails source consistency unless Q actually reports zero. Preserve measured zero rather than treating it as missing.

Guided: name the layers passed and explain why a schema-valid zero still needs source support.

Practise with less help: Audit types and fields

Use workbook W2 for the connected example “Structured outputs and validation”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Structured outputs and validation. Proposed explanation: “Valid JSON means correct answer”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which object demonstrates that valid JSON and valid types are insufficient for truth?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → RUN → VERIFY → EXPLAIN

Validate six output cards for required fields, types, allowed values, and evidence support; explain each rejection.

Validate six output cards for required fields, types, allowed values, and evidence support; explain each rejection.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



CREATOR MASTERY

1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or evidence path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



VERIFIED STUDY BUDDY

Require answer, source snippet, citation, uncertainty, and next-question fields.

THIS WEEK'S CHECKPOINT Require answer, source snippet, citation, uncertainty, and next-question fields.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

Evidence for your project

Validate Study Buddy outputs before another program consumes them; retain source IDs and separate syntax/type/support failures.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT CREATOR PROOF

Request fields, then check missing and invalid values.

CHALLENGE Request fields, then check missing and invalid values.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 8 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Schema requires title text and pages number. Output is {"title":"River","pages":"80"}.

The question

Diagnose the shape error.

Worked explanation

Pages is a string, not a number. Use numeric 80. Structure validation does not verify the real page count.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Type correctness is not factual correctness

Evidence or stage · Value or result

Evidence or stage: Wrong type

Value or result: pages = "80" is text

Evidence or stage: Correct type

Value or result: pages = 80 is numeric

Evidence or stage: Remaining check

Value or result: Verify the real page count

Schema requires title text and pages number. Output is {"title":"River","pages":"80"}.

Diagnose the shape error.

Pages is a string, not a number. Use numeric 80. Structure validation does not verify the real page count.

14. Try a new case independently

Independent case: Validate new objects independently

Sources A approve room F and B register F capacity 9. Object X has supported,F,9,[A,B] with exactly the required fields. Y replaces 9 with "9". Z keeps integer 9 but names room G. Trace the first failing layer for each and explain the difference.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 8 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A required schema has title as text and count as an integer. The output is {"title":"Seeds","count":"three"}. Identify the error and give a valid replacement.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Require answer, source snippet, citation, uncertainty, and next-question fields.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

Schema requires title:text and pages:number. Output is {"title":"Cloud","pages":"80"}. Fix its structural error. Would the repaired object prove the book really has 80 pages?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.